

Hosting the Blockchain Network

Options, Costs & Production Strategy

A technical decision report for the Medichain hybrid architecture:
self-hosted private chain vs. managed RPC vs. public Polygon deployment.

Project	Medichain — Blockchain-secured medical records
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Date	01 July 2026
Scope	Access-control + audit-log contract (MediAccessControl)
Stack	Flutter · Supabase · IPFS (Pinata) · EVM smart contract

1 Executive Summary

This report captures a working session on how to host the blockchain layer of Medichain. Medichain uses a **hybrid architecture**: identity is handled by Supabase Auth, record metadata by the Supabase database, encrypted files by IPFS (Pinata), and only **one smart contract** — `MediAccessControl` — lives on-chain, handling access control and an immutable audit log. Because the on-chain footprint is tiny, hosting choices are flexible and inexpensive.

The discussion moved through three questions: (1) what does it cost to host a **private** network, (2) what is the **best** way to host it, and (3) given the goal is simply to **host the contract** reliably and fast, what is the right path.

Bottom line

A smart contract is not something you host on a server — once deployed, the **chain hosts it**, replicated across every node automatically.

The public Amoy testnet felt slow because of a shared, rate-limited **public RPC endpoint** — not the chain itself.

Two viable production paths, and they trade off against each other: **zero infrastructure** (public chain + managed RPC, but per-transaction gas) OR **zero gas fees** (self-hosted private chain, but you run nodes).

For **host the contract only**: deploy to **Polygon** + a managed RPC (Alchemy/Infura) and sponsor user gas so patients never touch crypto.

2 Context: The Medichain On-Chain Footprint

Understanding what actually runs on-chain is what makes hosting cheap and simple. The responsibilities are split as follows:

Layer	Handled by	On-chain?
Identity / auth	Supabase Auth	No
Record metadata	Supabase Database	No
Encrypted files	IPFS (Pinata)	No
Access control (grant/revoke)	MediAccessControl contract	Yes
Audit log (view/download)	Contract events (immutable)	Yes

Compliance boundary — keep it intact

No PHI (patient health information) ever goes on-chain. On-chain data is immutable and cannot be deleted, which conflicts with the patient right-to-erasure. Files stay on IPFS, metadata in Supabase; only access-control entries and audit events are on-chain. This also keeps the blockchain nodes arguably out of HIPAA scope — the compliance burden stays with Supabase and IPFS.

3 Hosting Options & Costs

Three broad families of options were evaluated. The table summarises the trade-offs; detailed cost breakdowns follow.

Option	You run...	Gas fees	Monthly cost
A. Self-hosted private chain (Hyperledger Besu, QBFT)	The nodes	None (free gas)	~\$15 – 230
B. Public chain + managed RPC (Polygon + Alchemy/Infura)	Nothing	Tiny, per-tx	~\$0 + gas
C. Managed private chain (Kaleido, SettleMint)	Nothing	None	~\$300 – 1,000+

3.1 Self-hosted private chain (Besu) — cost tiers

Tier	Setup	Fault tolerant?	Monthly
Pilot / single node	1 VPS (2 vCPU / 4 GB)	No	~\$15 – 25
Production QBFT	4 validators + 1–2 RPC nodes	Yes (1 node)	~\$115 – 230

A 4-validator Byzantine-fault-tolerant (BFT) cluster tolerates one failing or malicious node (BFT needs $3f+1$ nodes). Free gas is achieved with `--min-gas-price=0` and `zeroBaseFee`.

3.2 Public chain (Polygon) — cost

- **Deployment:** one-time ~\$0.02 – 0.05 in gas.
- **Managed RPC (Alchemy/Infura):** \$0 — free tier (~300M units/month) easily covers Medichain's volume.
- **Per transaction:** ~\$0.001 – 0.005 each on Polygon (grant / revoke / view / download).

4 Production Architecture — Self-Hosted (Option A)

The production-grade design separates **consensus** from the **public surface**. Validators stay private and never face the internet; dedicated non-validator RPC nodes serve the app behind a load balancer and Cloudflare.

Layer	What	Why it is the production-right call
4 validators	QBFT, private, P2P only	Consensus survives 1 node loss; no RPC exposed to the internet.
2 RPC nodes	Non-validator full nodes	App entry point; if one dies the other serves. Validators stay isolated.
Cloudflare	TLS · WAF · DDoS · rate-limit	Hides origin IPs, blocks abuse, caps request rates, allowlists eth_* methods.
Load balancer	CF LB or nginx upstream	Health-checks RPC nodes; automatic failover.
Monitoring	Prometheus + Grafana + Alertmanager	Alerts on block-production stall, low peer count, disk full, validator down.
Backups / DR	Off-site keys + volume snapshots	Losing all keys = unrecoverable chain; keep keys backed up off-box.
IaC	Terraform + Ansible / compose	Reproducible, auditable, no snowflake servers.

Resilience recommendation

Spread nodes across **2 providers and 2+ regions** (e.g. Hetzner + Vultr). This survives a regional outage *and* a provider account suspension — a real risk you do not want killing a medical product. Production cost lands at **~\$70 – 140/month** including RPC nodes, monitoring and backups.

5 Fast Demo Setup (Single Besu Node)

For a product-owner demo the full cluster is unnecessary. A single Besu QBFT node on a small VPS is genuinely *blockchain-hosted*, fast (~2s blocks, instant confirmations, no public-RPC contention), and free-gas — while using the same chain ID and contract as the eventual production cluster, so nothing is thrown away.

Concern	How the single node delivers
"Must be blockchain hosted"	Real Hyperledger Besu on a VPS with a domain — genuinely hosted, not localhost.
"Amoy is too slow"	Dedicated node: ~2s blocks, instant confirmations, no rate limits.

Setup is ~30–45 minutes: generate genesis + key, `docker compose up -d` on a \$5 VPS (or locally with a Cloudflare tunnel), deploy the contract, then point the app's three constants at it.

6 Recommended: Host the Contract Only on Polygon

Since the goal is simply to host the contract (no infrastructure), the answer is to deploy to **Polygon** and reach it through a **managed RPC**. The repository is already wired for this — the `polygon` and `amoy` networks exist in `hardhat.config.js` and the `deploy` script supports them.

6.1 The four steps

- 1 **Get a fast RPC** — sign up at Alchemy, create a Polygon Mainnet (or Amoy) app, copy the HTTPS URL. This is what fixes the slowness.
- 2 **Fill `blockchain/.env`** — `PRIVATE_KEY`, `POLYGON_MAINNET_RPC_URL`, `POLYGONSCAN_API_KEY` (optional).
- 3 **Fund the deployer** — ~\$2 of POL on mainnet (one-time), or free faucet POL on Amoy.
- 4 **Deploy** — `npx hardhat run scripts/deploy.js --network polygon`, then paste the RPC URL, chain ID (137 mainnet / 80002 Amoy) and contract address into `contract_service.dart`.

6.2 Amoy (testnet) vs Mainnet

	Polygon Amoy (testnet)	Polygon Mainnet
Cost	Free (faucet gas)	Tiny real gas (~\$0.001/tx)
Speed (with Alchemy)	Fast	Fast
Permanence	Can be reset / deprecated	Permanent
Use for	Demo / staging	Real production

The one catch — who pays gas

On a public chain, every write (grantAccess, revokeAccess, logView, logDownload) is signed by the **user's wallet**, so each wallet needs a little POL. On Amoy you faucet it; on mainnet that is real money per user. For a clean medical-app UX, add **gas sponsorship** — an ERC-2771 meta-transaction relayer / paymaster (Biconomy, OpenZeppelin Defender, or ERC-4337 account abstraction) so patients never touch crypto.

7 Decision Guide

The fundamental trade-off cannot be avoided: you cannot get zero infrastructure *and* zero per-transaction gas at the same time.

If your priority is...	Choose	You accept
Zero infrastructure to run	Polygon + Alchemy + gas sponsorship	Small per-tx gas (sponsored for users)
Zero per-transaction gas fees	Self-hosted private Besu chain	You run and monitor the nodes
A fast, convincing demo soon	Amoy + Alchemy RPC, or single Besu node	Testnet may reset (Amoy)
No ops capacity, have budget	Managed private chain (Kaleido)	\$300–1,000+/month

Recommendation for Medichain

Given a solo / small team, cost-consciousness, and the goal of simply hosting the contract: **deploy to Polygon and reach it via Alchemy (managed RPC), with gas sponsorship** so users never need POL. It is production-ready, requires no servers, and directly resolves the public-testnet slowness. Keep the self-hosted 4-node Besu cluster as the fallback if zero gas fees ever becomes a hard requirement.

8 Repository Artifacts Created

During the session the following scaffolding was added to the repository to support the self-hosted route (kept available regardless of the final choice):

Path	Purpose
blockchain/besu/qbftConfigFile.json	4-validator QBFT network generator config (free gas, chain ID 131419)
blockchain/besu/docker-compose.yml	Per-node container definition for each of the 4 VPS
blockchain/besu/.env.example	Per-node settings (public IP, bootnodes, CORS)
blockchain/besu/README.md	Full 6-phase production runbook
blockchain/besu/demo/*	Single-node demo setup (config, compose, README)
blockchain/hardhat.config.js	Added a 'besu' network (free gas, env-driven RPC/chainId)
blockchain/scripts/deploy.js	Deploy script now resolves the Besu RPC endpoint
blockchain/.gitignore	Protects generated node keys and .env files from commits

Report generated from the Medichain blockchain-hosting consultation session. Figures are indicative estimates as of July 2026; verify current provider pricing and gas rates before committing.